

ANALYSIS QUALIFYING EXAM AUGUST 2025

Answer each question on a separate sheet of paper. Write solutions in a neat and logical fashion, giving complete reasons for all steps.

DO SIX OF THE EIGHT PROBLEMS.

- (1) (Do two. Short answer, you do not need to give a detailed proof, just a brief explanation.)

(a) If $a_1, \dots, a_n$ are non-negative numbers, must it be the case that

$$\left(\sum_{j=1}^n a_j \right)^5 \leq n^4 \sum_{j=1}^n a_j^5?$$

(b) Define $f \in L^1(\mathbb{R})$ by $f(x) = e^{-x^4} \chi_{[1, \infty)}(x)$, where χ denotes indicator function. Is the Fourier transform of f in $L^1(\mathbb{R})$?

(c) If $g, h \in L^1(\mu)$, when are $g d\mu$ and $h d\mu$ mutually singular?

- (2) Suppose $(X, \mathcal{M})$ is a measurable space. Show, if $E \in \mathcal{M} \otimes \mathcal{M}$, then

$$\{x \in X : (x, x) \in E\} \in \mathcal{M}.$$

- (3) (a) Let $f_n(x) = x^{-n}$ for $n \geq 2$. Does the numerical sequence

$$c_n = \int_1^\infty e^{inx} f_n(x) dx$$

converge?

(b) Give an example, if possible, where strict inequality holds in Fatou's Lemma.

- (4) Suppose X is a Banach space, $Y, Z \subseteq X$ are (closed) subspaces and $Y \cap Z = \{0\}$. Prove or disprove: there is a $C > 0$ such that

$$C(\|y\| + \|z\|) \leq \|y + z\|$$

for all $y \in Y$ and $z \in Z$.

- (5) Highlighting the key points, outline a proof of the following statement. If a sequence of functions converges in L^1 , then it has a subsequence that converges pointwise almost everywhere.
- (6) Suppose $(X, \mathcal{M}, \mu)$ is a measure space, $1 \leq p, q \leq \infty$ and $h : X \rightarrow \mathbb{C}$ is measurable. Show, if $hf \in L^q(\mu)$ for each $f \in L^p(\mu)$, then the map

$$L^p(\mu) \ni f \mapsto hf \in L^q(\mu)$$

is continuous.

- (7) Define the Hardy-Littlewood maximal function of a Lebesgue measurable function $f : \mathbb{R} \rightarrow \mathbb{R}$.

State the Hardy-Littlewood maximal inequality.

Show, if $f \in L^1(\mathbb{R})$, then f^* (the Hardy-Littlewood maximal function of f) is finite almost everywhere.

- (8) (Do two. Short answer, you do not need to give a detailed proof, just a brief explanation.)
- (a) Does there exist a bounded linear functional $\lambda : L^\infty(\mathbb{R}) \rightarrow \mathbb{C}$ for which there does *not* exist a $g \in L^1(\mathbb{R})$ such that $\lambda(f) = \int fg \, dx$?
- (b) Does the norm in $\ell^1(\mathbb{N})$ arise from an inner product?
- (c) Can a sigma algebra be bijective to $\mathbb{N}$?